Tilak Poudel

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PROFESSIONAL SUMMARY

Strategic **Full-Stack Software Engineer** with 5+ years of experience in architecting secure backend systems, responsive frontends, and **Generative AI integrations**. Expert in **Relational Database Design** and system optimization, currently pursuing an **M.Sc. in Data Science** to bridge the gap between Software Engineering and advanced data analytics. Proven track record in delivering scalable, secure solutions using modern DevOps and Agile methodologies.

TECHNICAL SKILLS

- **Full-Stack Mastery:** Node.js, PHP (Laravel), **TypeScript**, JavaScript (Vue.js, React), **Streamlit**, Livewire.
- **Engineering Practices: Repository Pattern**, Secure API Design, **ADRs**, Unit/Integration Testing, Feature Flags, Full SDLC.
- **Data & Databases: PostgreSQL**, **Prisma ORM**, SQL Optimization, MySQL, MongoDB, Python (Pandas, NumPy), **Scikit-learn**.
- **AI & Cloud: Generative AI (Gemini API)**, LLM Integration, Docker, CI/CD, Rate Limiting, Brute-force Protection.
- **Management:** Agile Scrum, Requirement Gathering, Jira, Trello, Technical Mentorship.

PROFESSIONAL EXPERIENCE

Part-Time Lecturer | Sahid Smarak College and Academia International College **2026 – Present**

- Teach UI/UX design, mobile programming, and machine learning through practical, real-world projects.

Workshop Facilitator: ML-Powered Web Application Development | Sahid Smarak College, Kathmandu **June 2026**

- Designed and delivered a 7-day intensive workshop in Python fundamentals, machine learning, and web application development.
- Trained 20+ students to build ML-powered applications using Python, Pandas, Scikit-learn, and Flask.
- Guided project teams building student performance prediction systems, achieving R^2 scores above 0.85.

Full Stack Developer (Freelance) | Aval Consulting Group (UK Client) **Feb 2026 – Apr 2026**

- Delivered a Civil Survey Management System from requirements gathering through production deployment.
- Integrated **Google Gemini AI** to automate professional report synthesis and text generation.
- Designed a normalized **PostgreSQL** schema with **Prisma ORM**, migrated legacy data, and implemented a maintainable Node.js/Express repository pattern in TypeScript.

Software Engineer | DzangoLab **2021 – Present**

- Improved database query performance by 30% through indexing and query optimization.
- Implemented secure **mTLS authentication** across microservices and feature flags for safe, iterative releases.
- Built Docker-based **CI/CD** pipelines that reduced deployment downtime by 40%.
- Delivered dashboards and data analytics for Wenow's carbon-footprint tracking solutions.

Lecturer & Trainer | Nepathya College **2019 – 2025**

- Taught Algorithms, DBMS, and programming to 300+ undergraduate students; designed practical curricula and mentored student projects and career development.

Full Stack Developer | Bootwal R&D **2018 – 2021**

- Developed full-stack applications and RESTful APIs with authentication for research and development projects.
- Collaborated with stakeholders to plan and build an employee management system for Lumbini Province, supporting activity tracking, promotion evaluation, and transfer recommendations.

PROJECTS

- **ChandaGyan: Sanskrit Verse Correction & Meter Identification (Ongoing MDS Thesis):** Proposed an NLP system that combines Sanskrit meter analysis with a context-aware masked language model to correct noisy Sanskrit verses and suggest likely meters. **Tribhuvan University** | Python, NLP, SanskritBERT.
- **Community Forest Management Web App:** Designed and developed a system for community forest operations, including stock, loans and debts, events, and billing. PHP (CodeIgniter), MySQL, jQuery, DataTables.
- **Lumbini Province Employee Management System:** Designed and developed a system to track employee activities, define promotion evaluation criteria, and recommend suitable office transfers based on employee history.
- **Customer Segmentation via Fuzzy Clustering:** Engineered a Streamlit dashboard applying **Fuzzy C-Means (FCM)** and RFM analysis to identify nuanced consumer segments. Implemented model comparison (FCM vs. K-Means) to optimize marketing precision. [GitHub] | [App] | [Medium]
- **Sentiment Analysis on COVID-19 Dataset:** Developed a Python pipeline to extract and classify public sentiment trends using NLP.
- **Social Network Analysis of PDF Texts:** Mapped thematic and philosophical networks across 108 Bhagavad Gita slokas and purports.
- **Data Analyst (Intern):** Conducted Trend Analysis and EDA on agricultural data, improving planning accuracy by 15%.

CERTIFICATIONS

- **Python and Statistics for Financial Analysis** | Coursera
- **Data Science Math Skills** | Duke University
- **Elements of AI** | University of Helsinki
- **Learning Linux Command Line** | LinkedIn Learning

VOLUNTEER WORK

Youth Preacher | ISKCON Nepal **2020 – Present**

- Organized 50+ seminars on various topics including time management, healthy habits, personal growth for 1000+ students.
- Developed website for ISKCON Tilottama Gaushala.

EDUCATION

- **Master of Science in Data Science (Ongoing)** | Tribhuvan University **2025 – Present**
- **BSc. CSIT** | Nepathya College (Graduated with Distinction) **2015 – 2019**